EX.NO.: 13
DATE: 18.03.2025

TEXT TO VIDEO GENERATION

AIM
To generate short video clips from textual descriptions using a Transformer-based diffusion model.

DATASET DESCRIPTION
- The MSR-VTT dataset is a large-scale benchmark for video understanding, containing **10,000 video clips and 200,000 textual captions**, covering diverse scenes, actions, and objects.

TECHNIQUES USED
Diffusion Model Architecture:
Utilizes **Damo-Vilab's Text-to-Video Diffusion Model (MS-1.7B)**, a state-of-the-art model that leverages **Transformer-based denoising diffusion probabilistic models (DDPMs)** for high-quality video generation.

Diffusion Pipeline:
The pipeline leverages latent diffusion techniques to progressively refine noise into coherent video frames.

Frame Stitching Approach:
To enhance the number of frames, the pipeline is executed multiple times, and the frames are combined to produce a continuous video.

Torch CUDA Acceleration:
GPU-based computation for faster inference and high-quality generation.

CODE AND OUTPUT
```
!pip install torch diffusers accelerate

import torch
from diffusers import DiffusionPipeline

pipe = DiffusionPipeline.from_pretrained("damo-vilab/text-to-video-ms-1.7b",
torch_dtype=torch.float16, variant="fp16")
pipe = pipe.to("cuda")

prompt = "penguin dribbling and dunking basketball"
# Generate more frames by running the pipeline multiple times
num_iterations = 4  # Number of times to run the pipeline for more frames
all_frames = []

for _ in range(num_iterations):
    video_frames = pipe(prompt).frames[0]
    all_frames.extend(video_frames)

from diffusers.utils import export_to_video

video_path = export_to_video(all_frames)
print(f"Video saved at: {video_path}")

from google.colab import files

files.download(video_path)
```